

Discussion 5 Solutions

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October 2025

Problem 1: For each of the following sets V , determine whether V is a vector space with the operations of usual vector addition and scalar multiplication over $\mathbb{R}$. Justify your answer in each case.

(a) $V = \{(x, y) \in \mathbb{Z}^2\}$

Solution:

(1) Closure under addition

Let $(x_1, y_1), (x_2, y_2) \in V$. Then

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2).$$

Since $\mathbb{Z}$ is closed under addition, both $x_1 + x_2$ and $y_1 + y_2$ are integers. Hence $(x_1 + x_2, y_1 + y_2) \in V$. So V is closed under addition.

(2) Closure under scalar multiplication

Let $\alpha \in \mathbb{R}$ and $(x, y) \in V$. Then

$$\alpha(x, y) = (\alpha x, \alpha y).$$

If α is not an integer, αx and αy are generally **not integers**. For example, if $\alpha = \frac{1}{2}$ and $(x, y) = (2, 4)$,

$$\alpha(x, y) = (1, 2) \in \mathbb{Z}^2,$$

but if $(x, y) = (1, 1)$, then

$$\alpha(x, y) = \left(\frac{1}{2}, \frac{1}{2}\right) \notin \mathbb{Z}^2.$$

Therefore, V is not closed under scalar multiplication over $\mathbb{R}$. Hence,

V is not a vector space over $\mathbb{R}$.

(b) $V = \{(x, y, z, w) \in \mathbb{R}^4 \mid x + 2y - z + w = 5\}$

Solution:

(1) Closure under addition

Let $(x_1, y_1, z_1, w_1), (x_2, y_2, z_2, w_2) \in V$. Then

$$x_1 + 2y_1 - z_1 + w_1 = 5, \quad \text{and} \quad x_2 + 2y_2 - z_2 + w_2 = 5.$$

Their sum is

$$(x_1, y_1, z_1, w_1) + (x_2, y_2, z_2, w_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2, w_1 + w_2).$$

Compute:

$$(x_1 + x_2) + 2(y_1 + y_2) - (z_1 + z_2) + (w_1 + w_2) = (x_1 + 2y_1 - z_1 + w_1) + (x_2 + 2y_2 - z_2 + w_2) = 5 + 5 = 10 \neq 5.$$

Thus, $(x_1 + x_2, y_1 + y_2, z_1 + z_2, w_1 + w_2) \notin V$.

Therefore, V is not closed under addition.

V is not a vector space over $\mathbb{R}$.

Problem 2: In each of the following, V is a vector space and $S \subseteq V$. Determine whether S is a valid subspace of V under the same operations as V . Justify your answer in each case.

(a) $V = \mathbb{R}^2$ and S is the x-axis.

Solution:

$$V = \mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}, \quad S = \{(x, 0) \mid x \in \mathbb{R}\}.$$

(1) Presence of zero vector

The zero vector in V is $(0, 0)$. Since $(0, 0) = (x, 0)$ when $x = 0$, we have $(0, 0) \in S$. Hence, S contains the zero vector.

(2) Closure under addition

Let $(x_1, 0), (x_2, 0) \in S$. Then

$$(x_1, 0) + (x_2, 0) = (x_1 + x_2, 0).$$

Since $x_1 + x_2 \in \mathbb{R}$, $(x_1 + x_2, 0) \in S$. Thus, S is closed under addition.

(3) Closure under scalar multiplication

Let $\alpha \in \mathbb{R}$ and $(x, 0) \in S$. Then

$$\alpha(x, 0) = (\alpha x, 0).$$

Since $\alpha x \in \mathbb{R}$, $(\alpha x, 0) \in S$. Hence, S is closed under scalar multiplication.

All subspace properties are satisfied. Therefore,

$$\boxed{S \text{ is a subspace of } V = \mathbb{R}^2.}$$

- (b) $V = \mathbb{R}^3$ and $S = \{(x, y, z) \in \mathbb{R}^3 \mid x - 2y + z = 1\}$.

Solution:

$$V = \mathbb{R}^3 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}, \quad S = \{(x, y, z) \mid x - 2y + z = 1\}.$$

- (1) Presence of zero vector

The zero vector in V is $(0, 0, 0)$. Check if it belongs to S :

$$0 - 2(0) + 0 = 0 \neq 1.$$

Thus, $(0, 0, 0) \notin S$.

Since S does not contain the zero vector, it cannot be a subspace of $\mathbb{R}^3$.

$$\boxed{S \text{ is not a subspace of } \mathbb{R}^3.}$$

- (c) $V = \mathbb{R}^2$ with vector addition and scalar multiplication defined by

$$(x_1, y_1) \oplus (x_2, y_2) = (x_1 + x_2, y_1 + y_2 - 5), \quad \alpha \odot (x, y) = (\alpha x, \alpha y + (1 - \alpha)5),$$

and $S = \{(0, y) \mid y \in \mathbb{R}\}$.

Solution:

$$V = \mathbb{R}^2, \quad S = \{(0, y) \mid y \in \mathbb{R}\}.$$

- (1) Presence of zero vector

The zero vector in V (with the given operations) is $(0, 5)$. Since $(0, 5) = (0, y)$ when $y = 5$, we have $(0, 5) \in S$. Hence, S contains the zero vector.

- (2) Closure under addition

Let $(0, y_1), (0, y_2) \in S$. Then

$$(0, y_1) \oplus (0, y_2) = (0 + 0, y_1 + y_2 - 5) = (0, y_1 + y_2 - 5).$$

Since $y_1 + y_2 - 5 \in \mathbb{R}$, the result is of the form $(0, y)$, so $(0, y_1) \oplus (0, y_2) \in S$. Hence, S is closed under addition.

(3) Closure under scalar multiplication

Let $\alpha \in \mathbb{R}$ and $(0, y) \in S$. Then

$$\alpha \odot (0, y) = (\alpha \cdot 0, \alpha y + (1 - \alpha)5) = (0, \alpha y + (1 - \alpha)5).$$

Since $\alpha y + (1 - \alpha)5 \in \mathbb{R}$, the result is again of the form $(0, y')$. Therefore, $\alpha \odot (0, y) \in S$.

All subspace properties are satisfied. Therefore,

S is a subspace of V .

Problem 3: A superhero is at the origin of a two-dimensional plane. The superhero has two devices: a hoverboard that moves any distance in the direction $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and a magic carpet that moves any distance in the direction $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

- (a) An evil villain is hiding out on the plane at the point $\begin{pmatrix} -5 \\ 7 \end{pmatrix}$. How many hoverboard units and magic carpet units does the superhero have to move to get to the villain?
- (b) Is there anywhere in the plane that the villain could safely hide and not be reached?
- (c) The superhero and the villain are transported to a three-dimensional space where the superhero now has three devices:

$$\text{hoverboard: } \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \quad \text{magic carpet: } \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \quad \text{scooter: } \begin{pmatrix} -2 \\ 0 \\ 6 \end{pmatrix}$$

Is there anywhere that the villain could safely hide?

Solution:

- (a) We want scalars $a, b \in \mathbb{R}$ such that

$$a \begin{pmatrix} 3 \\ 1 \end{pmatrix} + b \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -5 \\ 7 \end{pmatrix}.$$

Equivalently,

$$\begin{cases} 3a + b = -5, \\ a + 2b = 7. \end{cases}$$

Form the augmented matrix and apply Gauss–Jordan elimination:

$$\begin{aligned}
& \left[\begin{array}{cc|c} 3 & 1 & -5 \\ 1 & 2 & 7 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|c} 1 & 2 & 7 \\ 3 & 1 & -5 \end{array} \right] \\
& \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & -5 & -26 \end{array} \right] \xrightarrow{R_2 \rightarrow \frac{1}{-5}R_2} \left[\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & 1 & \frac{26}{5} \end{array} \right] \\
& \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left[\begin{array}{cc|c} 1 & 0 & 7 - 2\left(\frac{26}{5}\right) \\ 0 & 1 & \frac{26}{5} \end{array} \right] = \left[\begin{array}{cc|c} 1 & 0 & \frac{35-52}{5} \\ 0 & 1 & \frac{26}{5} \end{array} \right] = \left[\begin{array}{cc|c} 1 & 0 & -\frac{17}{5} \\ 0 & 1 & \frac{26}{5} \end{array} \right].
\end{aligned}$$

Hence,

$$a = -\frac{17}{5}, \quad b = \frac{26}{5}.$$

Therefore, the superhero must move $-\frac{17}{5}$ hoverboard units and $\frac{26}{5}$ magic carpet units to reach the villain.

- (b) To determine if there is anywhere in the plane the villain could safely hide, we check whether

$$\vec{v}_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad \vec{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

span all of $\mathbb{R}^2$.

We form the matrix with these vectors as columns:

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}.$$

Apply Gauss–Jordan elimination to check if the matrix reduces to the identity.

$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - \frac{1}{3}R_1} \begin{bmatrix} 3 & 1 \\ 0 & 2 - \frac{1}{3} \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & \frac{5}{3} \end{bmatrix} \xrightarrow{R_1 \rightarrow \frac{1}{3}R_1, R_2 \rightarrow \frac{3}{5}R_2} \begin{bmatrix} 1 & \frac{1}{3} \\ 0 & 1 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - \frac{1}{3}R_2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

This is the identity matrix, which means the two direction vectors $\vec{v}_1$ and $\vec{v}_2$ are linearly independent.

Since the two directions span $\mathbb{R}^2$, the villain cannot hide anywhere in the plane.

- (c) Now the superhero and villain are in $\mathbb{R}^3$, and the superhero has three devices:

$$\vec{v}_1 = \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}, \quad \vec{v}_2 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{v}_3 = \begin{pmatrix} -2 \\ 0 \\ 6 \end{pmatrix}.$$

We check whether $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are linearly independent, i.e., whether they span all of $\mathbb{R}^3$.

Form the matrix with these vectors as columns:

$$A = \begin{bmatrix} -1 & 2 & -2 \\ 0 & 1 & 0 \\ 3 & 0 & 6 \end{bmatrix}.$$

We will apply Gauss–Jordan elimination.

$$\begin{aligned} \begin{bmatrix} -1 & 2 & -2 \\ 0 & 1 & 0 \\ 3 & 0 & 6 \end{bmatrix} &\xrightarrow{R_3 \leftarrow R_3 + 3R_1} \begin{bmatrix} -1 & 2 & -2 \\ 0 & 1 & 0 \\ 0 & 6 & 0 \end{bmatrix} \xrightarrow{R_1 \leftarrow -R_1} \begin{bmatrix} 1 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 6 & 0 \end{bmatrix} \\ &\xrightarrow{R_3 \leftarrow R_3 - 6R_2} \begin{bmatrix} 1 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \leftarrow R_1 + 2R_2} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \end{aligned}$$

The resulting matrix has two pivots (in the first and second columns).

Therefore, the three vectors are linearly dependent ($\vec{v}_3 = 2\vec{v}_1$) and span only a plane in $\mathbb{R}^3$, not the entire space. The villain can hide anywhere, not lying on this plane.

Problem 4: Identify whether each subset is linearly dependent or linearly independent and if they form a basis for $\mathbb{R}^3$.

(a) $\left\{ \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} \right\}$

Solution: Let these vectors be $\vec{v}_1, \vec{v}_2$. To test for linear independence, consider the homogeneous system

$$a\vec{v}_1 + b\vec{v}_2 = \vec{0}.$$

This gives the augmented matrix

$$\left[\begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 0 & 0 \\ -1 & 4 & 0 \end{array} \right].$$

Apply Gauss–Jordan elimination.

$$\left[\begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 0 & 0 \\ -1 & 4 & 0 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{cc|c} -1 & 4 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] \xrightarrow{R_1 \leftarrow -R_1} \left[\begin{array}{cc|c} 1 & -4 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] \xrightarrow{R_1 \leftarrow R_1 + 4R_3} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right].$$

Therefore, $a = 0, b = 0$. Thus, the two vectors are linearly independent.

However, because there are only two vectors in $\mathbb{R}^3$, they span only a plane, not the entire space. Hence, they do not form a basis for $\mathbb{R}^3$.

(b) $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$

Solution: Let these vectors be $\vec{v}_1, \vec{v}_2, \vec{v}_3$. To test for linear independence, consider the homogeneous system

$$a\vec{v}_1 + b\vec{v}_2 + c\vec{v}_3 = \vec{0}.$$

This gives the augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ 3 & 1 & 1 & 0 \end{array} \right].$$

Apply Gauss–Jordan elimination.

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ 3 & 1 & 1 & 0 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1, R_3 \leftarrow R_3 - 3R_1} \left[\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & -8 & 1 & 0 \end{array} \right]$$

$$\xrightarrow{R_3 \leftarrow R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \xrightarrow{R_2 \leftarrow -\frac{1}{4}R_2} \left[\begin{array}{ccc|c} 1 & 3 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \xrightarrow{R_1 \leftarrow R_1 - 3R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right].$$

Therefore, $a = 0, b = 0, c = 0$. Hence, the vectors are linearly independent and form a basis for $\mathbb{R}^3$.

(c) $\left\{ \begin{pmatrix} 9 \\ 9 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 5 \\ -4 \end{pmatrix}, \begin{pmatrix} 12 \\ 12 \\ -1 \end{pmatrix} \right\}$

Solution: In $\mathbb{R}^3$, there can be at most three linearly independent vectors. Therefore, at least one of the given four vectors can be expressed as a linear combination of the others. Hence, the set is linearly dependent and does not form a basis for $\mathbb{R}^3$.

Problem 5: Represent the vector $\vec{v}$ with respect to each of the two bases.

$$\vec{v} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \quad B_1 = \left\langle \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\rangle, \quad B_2 = \left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \end{pmatrix} \right\rangle.$$

Solution:

(a) Coordinates with respect to B_1

We want scalars a, b such that

$$a \begin{pmatrix} 1 \\ -1 \end{pmatrix} + b \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

This gives the system

$$\begin{cases} a + b = 3, \\ -a + b = -1. \end{cases}$$

Form the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ -1 & 1 & -1 \end{array} \right].$$

Apply Gauss–Jordan elimination.

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ -1 & 1 & -1 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 + R_1} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 2 & 2 \end{array} \right] \xrightarrow{R_2 \leftarrow \frac{1}{2} R_2} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & 1 \end{array} \right] \xrightarrow{R_1 \leftarrow R_1 - R_2} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array} \right].$$

Hence, the new coordinates are

$$a = 2, \quad b = 1.$$

(b) Coordinates with respect to B_2

We want scalars c, d such that

$$c \begin{pmatrix} 1 \\ 2 \end{pmatrix} + d \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

This gives the system

$$\begin{cases} c + d = 3, \\ 2c + 3d = -1. \end{cases}$$

Form the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 3 & -1 \end{array} \right].$$

Apply Gauss–Jordan elimination.

$$\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & 3 & -1 \end{array} \right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & 1 & -7 \end{array} \right] \xrightarrow{R_1 \leftarrow R_1 - R_2} \left[\begin{array}{cc|c} 1 & 0 & 10 \\ 0 & 1 & -7 \end{array} \right].$$

Hence, the new coordinates are

$$c = 10, \quad d = -7.$$